

Iterative methods for solving linear systems

$$\underbrace{x^{(0)}}_{\text{initial guess}} \longrightarrow x^{(1)} \longrightarrow x^{(2)} \longrightarrow \dots x^{(k)}$$

$$\lim_{k \rightarrow \infty} \|x - x^{(k)}\| = 0$$

$$Ax = b \quad x = A^{-1}b$$

$$\det(A) \neq 0$$

Linear iterative methods

$$x^{(k+1)} = Bx^{(k)} + f$$

$$x = Bx + f \iff Ax = b \iff x = A^{-1}b$$

$$A^{-1}b = BA^{-1}b + f \iff$$

$$\boxed{\begin{aligned} f &= A^{-1}b - BA^{-1}b \\ &= (I - B)A^{-1}b \end{aligned}}$$

consistency

Convergence

Sufficient condition $\|B\| < 1$

Sufficient and necessary condition $\rho(B) < 1$

$$\rho(B) = \max_j |\lambda_j(B)|$$

$$\rho(B) = \inf_p \|B\|_p \Rightarrow \rho(B) \leq \|B\|_p \quad \forall p$$

$$\text{if } B \text{ is SPD } \rho(B) = \|B\|_2$$

Jacobi and Gauss-Seidel methods

$$A = L + D + U$$



$$Ax = b$$

$$(L + D + U)x = b$$

$$Dx = -(L + U)x + D^{-1}b$$

$$x = -D^{-1}(L + U)x + D^{-1}b$$

$$x^{(k+1)} = -D^{-1}(L + U)x^{(k)} + D^{-1}b \quad \text{Jacobi}$$

$$B_J = -\underline{D}^{-1}(L+U) \quad f_J = \underline{D}^{-1}b$$

$$(L+D+U)x = b$$

$$(L+D)x = -Ux + b$$

$$x = -(L+D)^{-1}Ux + (L+D)^{-1}b$$

Gauß-Seidel

$$x^{(k+1)} = -(L+D)^{-1}Ux^{(k)} + (L+D)^{-1}b$$

$$B_{GS} = -\underline{(L+D)^{-1}U} \quad f_{GS} = \underline{(L+D)^{-1}b}$$

Specific conditions for convergence of J. and G.S.

- If the matrix A is strictly diagonally dominant by rows $\Rightarrow \|B_J\|_1 < 1$
 $\Rightarrow$ convergence
 $\|B_{GS}\|_1 < 1$

- If the matrix is strictly diagonally dominant by columns

$$\Rightarrow \|B_J\|_\infty < 1$$

$$\|B_{GS}\|_\infty < 1 \Rightarrow \text{both methods will converge}$$

• If $A \Rightarrow \text{SPD} \Rightarrow \text{GS}$ will converge

• If A is tridiagonal and SPT,
 $\Rightarrow$ both J and GS
will converge and
 GS will converge in
fewer iterations

tridiagonal: $A = [a_{ij}] \quad a_{ij} = 0 \quad \text{if } |i-j| > 1$

Richardson method (stationary)

$$x^{(k+1)} = x^{(k)} + \underbrace{(\quad)}_{=0 \text{ if } x^{(k)} = A^{-1}b}$$

$$r^{(k)} = b - Ax^{(k)} \quad k\text{-th residual}$$

$$x^{(k+1)} = x^{(k)} + \alpha r^{(k)}$$

Stationary
Richardson
method

α does not depend on k
 $\Rightarrow$ stationary method

$$x^{(k+1)} = x^{(k)} + \alpha (b - Ax^{(k)})$$

$$x^{(k+1)} = (I - \alpha A) x^{(k)} + \alpha b$$

$$= \underline{\underline{B_{SR}(\alpha)}} x^{(k)} + \underline{\underline{f_{SR}(\alpha)}}$$

From now on we assume A is SPD

$$\Rightarrow \lambda_i(A) > 0 \quad \forall i$$

$$\lambda_{\max}(A) = \lambda_1(A) \geq \lambda_2(A) \geq \lambda_3(A) \cdots \geq \lambda_n(A) = \lambda_{\min}(A) > 0$$

Theorem If A is SPD the stationary Richardson method will be convergent if and only if

$$0 < \alpha < \frac{2}{\lambda_{\max}(A)}$$

$\mu_i(B)$ eigenvalues of B

$$B = I - \alpha A$$

$$Bx = \mu_i x$$

$$(I - \alpha A)x = \mu_i x$$

$$x - \alpha Ax = \mu_i x$$

$$Ax = \lambda_i x$$

$$x - \alpha \lambda_i x = \mu_i x$$

$$(1 - \alpha \lambda_i) = \mu_i$$

$$|\mu_i| < 1 \quad |1 - \alpha \lambda_i| < 1$$

$$-1 < 1 - \alpha \lambda_i < 1$$

$$1 > \alpha \lambda_i, -1 > -1$$

$$2 > \alpha \lambda_i > 0$$

$$\boxed{\alpha > 0}$$

$$\frac{2}{\lambda_i} > \alpha \Leftarrow$$

$$\boxed{\frac{2}{\lambda_{\max}} > \alpha}$$

A - wozu

$$A \in \mathbb{R}^{n \times n}$$

A SPD

$$\|z\|_A = z^T A z \geq 0 \quad (=0 \text{ iff } z=0)$$

$$\|e^{(k+1)}\|_A \leq \rho(B_{SR}(\alpha)) \|e^{(k)}\|_A$$

the convergence rate of S. Richardson will be faster if $\rho(B_{SR}(\alpha))$ is minimized

$$\alpha_{\text{opt}} = \underset{0 < \alpha < \frac{2}{\lambda_{\max}}}{\text{argmin}} \left\{ \max_i |1 - \alpha \lambda_i(A)| \right\}$$

$$\alpha_{\text{opt}} = \frac{2}{\lambda_{\min}(A) + \lambda_{\max}(A)}$$

optimal parameter

if A is SPD then $\|A\|_2 = \lambda_{\max}(A)$

$$\begin{aligned} \rho_{\text{opt}} = \rho(B_{SR}(\alpha_{\text{opt}})) &= -1 + \alpha_{\text{opt}} \lambda_{\max}(A) = \\ &= 1 - \alpha_{\text{opt}} \lambda_{\min}(A) \\ &= \frac{\lambda_{\max}(A) - \lambda_{\min}(A)}{\lambda_{\max}(A) + \lambda_{\min}(A)} \end{aligned}$$

$$\lambda_i(A^{-1}) = \frac{1}{\lambda_i(A)} \Rightarrow$$

$$s_{\text{opt}} = \frac{\kappa_2(A) - 1}{\kappa_2(A) + 1}$$

- The convergence rate of the stationary Richardson method depends on the conditioning of A

"Preconditioned" Richardson method

$$P^{-1}Ax = P^{-1}b \quad \det(P) \neq 0$$

A is SPD

$P^{-1}A$ is SPD

$$x^{(k+1)} = x^{(k)} + \alpha (P^{-1}b - P^{-1}Ax^{(k)})$$

$$x^{(k+1)} = x^{(k)} + \alpha P^{-1}r^{(k)}$$

Preconditioned Richardson method

convergence

$$0 < \alpha < \frac{2}{\lambda_{\max}(P^{-1}A)}$$

optimal α

$$\alpha_{\text{opt}} = \frac{2}{\lambda_{\max}(P^{-1}A) + \lambda_{\min}(P^{-1}A)}$$

optimal ρ $\rho_{\text{opt}} = \frac{\kappa_2(P^{-1}A) - 1}{\kappa_2(P^{-1}A) + 1}$

so if $\kappa_2(P^{-1}A) \ll \kappa_2(A)$ we obtain a method that converges in fewer iterations

Example $P = A$ $P^{-1}A = I$

$$\begin{aligned} x^{(k+1)} &= x^{(k)} + \alpha P^{-1}(b - Ax^{(k)}) \\ &= x^{(k)} + \alpha P^{-1}b - \alpha P^{-1}Ax^{(k)} \\ &= x^{(k)} + \alpha x_{\text{opt}} - \alpha x^{(k)} \end{aligned}$$

$$\alpha_{\text{opt}} = \frac{2}{1} = 1 \quad = x$$

Reducing the number of iterations is, by itself, not sufficient to reduce the actual cost of solving the system

- We need P to be easy to construct
- We need $P^{-1}z$ be easy to compute for any z

Possible common choices of P

$$P = \text{diag}(\text{diag}(A))$$

$$x^{(k+1)} = x^{(k)} + \alpha D^{-1} (b - Ax^{(k)})$$

$$\alpha = 1$$

$$D x^{(k+1)} = D x^{(k)} + b - A x^{(k)}$$

$$= (D - A) x^{(k)} + b$$

$$= (D - L - D - U) x^{(k)} + b$$

$$= -(L + U) x^{(k)} + b$$

$$x^{(k+1)} = -D^{-1}(L + U) x^{(k)} + D^{-1}b$$

Jacobi method

$$P = \text{diag}(\text{diag}(A)) \quad \text{Jacobi preconditioner}$$

$$P = L + D \quad \text{Gauss-Seidel preconditioner}$$